

Prometheus: A mobile telepresence system connecting the 1st person and 3rd person perspectives continuously

Ryotaro Kimura
The University of Tokyo
QuantumCore Inc.
Tokyo, Japan
ryoppippi@g.ecc.u-tokyo.ac.jp

Jun Rekimoto
Sony CSL Kyoto
Kyoto, Japan
The University of Tokyo
Tokyo, Japan
rekimoto@acm.org

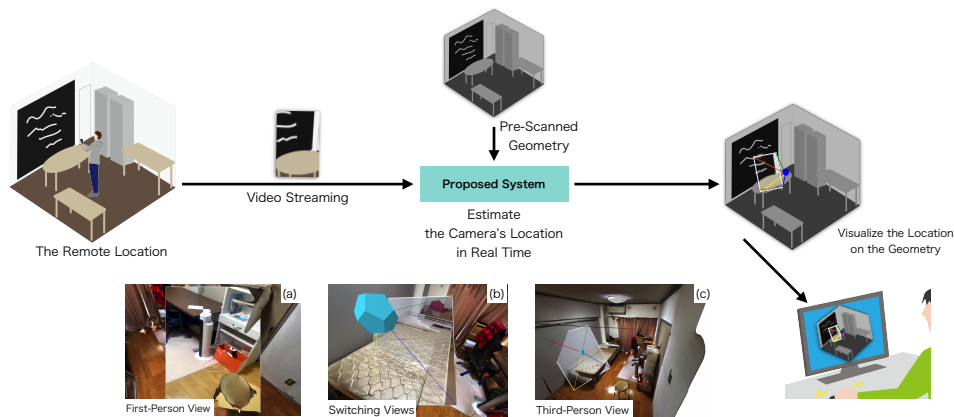


Figure 1: System overview. A remote worker captures the area with a camera and streams the image (we call it first-person view) to our system. Then, the system estimates the camera's position and direction from the image and the pre-scanned 3D geometry. Finally, our system generates a third-person view by overlapping the captured image and the location information on the geometry. The receiver can switch these views freely.

(a) First-person view: The image captured by remote worker is displayed in the center, with the pre-scanned geometry around it. This view moves with the movement of the remote worker. (b) Switching between first-person and third-person view: The receiver can switch between the two views. (c) Third-person view: The user can move the viewpoint freely in the geometry model. The first-person view and the camera's position are also displayed so that the receiver can know the location of the remote worker.

CCS CONCEPTS

• **Human-centered computing** → **Treemaps; Visualization design and evaluation methods; Collaborative interaction.**

KEYWORDS

Telepresence, Collaborative Interactions, Visualization, Real-World Capturing

ACM Reference Format:

Ryotaro Kimura and Jun Rekimoto. 2022. Prometheus: A mobile telepresence system connecting the 1st person and 3rd person perspectives continuously.

In *SIGGRAPH Asia 2022 Posters (SA '22 Posters)*, December 06-09, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3550082.3564187>

1 INTRODUCTION

The development of telepresence is underway. Local users can collaborate with remote workers by using cameras and other sensors worn to monitor the situation at the remote location [Minsky 1980]. Unlike two-dimensional teleconferencing, such as Zoom, telepresence allows remote workers to work while moving freely around the remote site.

A camera worn by a remote worker is used to monitor the case in the remote area. These images would transmit the *first-person view* of the remote workers.

However, the following issues arise with first-person view images alone.

First, the first-person views are not suitable for understanding the spatial structure and the current location in the space. The angle of view of the images depends on the direction in which the remote operator is looking, and the angle is often not wide enough to see the

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '22 Posters, December 06-09, 2022, Daegu, Republic of Korea

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9462-8/22/12.

<https://doi.org/10.1145/3550082.3564187>

entire space. When the receiver needs to get spatial information, it is necessary to move the viewpoint widely. However, the receiver of the images cannot freely control the direction of the images because the direction of the image depends on the worker's position and movement. Therefore, it is hard to grasp the entire space from a first-person view.

One way to solve these problems is to acquire and reconstruct the remote location as three-dimensional(3D) information in real-time. This real-time 3D geometry makes it possible to view the remote area from any viewpoint, not limited to the first-person's view(we call this view as *third-person view*). For example, an image from the remote operator's overhead viewpoint makes it easier to grasp the spatial position of the remote operator and the surrounding situation. Some studies use (multiple) depth sensors laid out in advance in the remote workspace [Komiya et al. 2017]. In other studies, mobile sensors are used to scan spacial information, such as handy depth sensors [Young et al. 2020], sensors on drones [Isop et al. 2019], and so on.

However, operating this 3D information has problems. First, the available space is restricted when we use the installed sensors in the remote area. It needs extra operations to use the sensors, including calibration, managing network connection, etc. Therefore, it is hard to expand the available space. In addition, even though we use a mobile camera, there is another issue: the reconstructed images are of poor quality compared to those from a regular camera. Finally, real-time reconstruction needs high computational resources. Also, streaming this information requires a large amount of data transfer.

2 OUR APPROACH

In this study, we propose a telepresence method that generates first- and third-person views in real-time by combining static spatial information measured in 3D beforehand with images from a monocular camera worn or held by a remote worker. Specifically, we utilized the VPS (Visual Positioning System) [Tilman Reinhardt and Software Engineer 2019] method to estimate the camera position from the first-person viewpoint video(Figure 1). Then, it superimposes the previously scanned geometry position to construct a pseudo-third-person viewpoint(as seen in Figure 1c). We also provide an interface that seamlessly switches between these two viewpoints. The receiver can simultaneously grasp real-time information about the site and its spatial structure by freely switching between the two views.

As we discussed, scanning spatial information in real-time is challenging, which is a significant obstacle to using viewpoint generation. On the other hand, the background buildings, walls, and ground in the space are static of the remote spatial information. It is, therefore, not always necessary to acquire these in real-time. The static pre-scanned geometry scanned has sufficient information to present the space's structure and estimate the camera's position. We utilized this geometry to estimate the camera's position.

There are several advantages to this method. First, it is easy to scale up the available area. In previous studies, the threshold for acquiring spatial information was high, as equipment had to be installed and calibrated in the space, which led to the problem that the available space was limited. In our method, all the user should do in advance is to scan the area to be used. As no calibration for

the sensors is needed, increasing the available space area is easy. In addition, if existing 3D maps are used, users can use the system by skipping the act of scanning the space. Furthermore, the equipment-independent nature of this system also makes it possible to use it outdoors.

Second, it is easy to increase the number of viewpoints. As this method uses only static geometry and camera image, it is not difficult to increase the number of views by introducing multiple cameras. In addition, as we use a regular camera, it is not challenging to increase the number of cameras. When there are multiple first-person views, the receiver can move across these views, making it easier to understand the space in remote areas (Figure 2).

Finally, the system can generate third-person viewpoint video images that are as good as the camera images by enhancing the quality of the pre-scanned geometry. The methods of generating high-quality models are currently being pioneered in photogrammetry technology. Therefore, this system is highly compatible with this field (Figure 2).

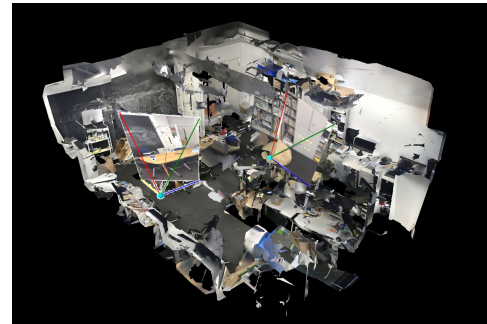


Figure 2: Example of Combining our prototype with another 3D model, which was scanned with MatterPort and Rico Theta Z1

ACKNOWLEDGMENTS

This work was supported by JST Moonshot R&D Grant Number JPMJMS2012, JST CREST Grant Number JPMJCR17A3, the commissioned research by National Institute of Information and Communications Technology (NICT) Japan, and The University of Tokyo Human Augmentation Research Initiative.

REFERENCES

- Werner Alexander Isop, Christoph Gebhardt, Tobias Nageli, Friedrich Fraundorfer, Otmar Hilliges, and Dieter Schmalstieg. 2019. High-Level Teleoperation System for Aerial Exploration of Indoor Environments. *Frontiers in Robotics and AI* 6 (2019). <https://doi.org/10.3389/frobt.2019.00095>
- Ryohei Komiya, Takashi Miyaki, and Jun Rekimoto. 2017. JackIn Space: Designing a Seamless Transition between First and Third Person View for Effective Telepresence Collaborations. In *Proceedings of the 8th Augmented Human International Conference (Silicon Valley, California, USA) (AH '17)*. Association for Computing Machinery, New York, NY, USA, Article 14, 9 pages. <https://doi.org/10.1145/3041164.3041183>
- Marvin Minsky. 1980. Telepresence. *OMNI* July (1980), 45–51.
- Tilman Reinhardt and Software Engineer. 2019. *Using Global Localization to Improve Navigation*. Retrieved Jan 13, 2022 from <https://ai.googleblog.com/2019/02/using-global-localization-to-improve.html>
- Jacob Young, Tobias Langlotz, Steven Mills, and Holger Regenbrecht. 2020. Mobileportation: Nomadic Telepresence for Mobile Devices. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 4, 2, Article 65 (jun 2020), 16 pages. <https://doi.org/10.1145/3397331>